```
#data used diamonds
View(diamonds)

install.packages("ggplot2")
library(ggplot2)
#Bar plot

#Map aesthetic
ggplot(diamonds, aes(cut))

# no need to write aesthetic x=  or y= , by default first variable
will be condered for x and second for y
ggplot(diamonds, aes(cut))

#Add geometric as bar by geom_bar
# for a univariate categorical variable
ggplot(diamonds, aes(x=cut))+geom_bar()

print(P)
# Use actual y-values from the data instead of counting rows
#flip the coordinate
ggplot(diamonds, aes(x=cut))+geom_bar()+coord_flip()

#fill colors to bar
ggplot(diamonds, aes(x=cut))+geom_bar(fill="red")
ggplot(diamonds, aes(x=cut))+geom_bar(fill="red", alpha=0.8)
# border color
ggplot(diamonds, aes(x=cut))+geom_bar(fill="red", color="blue")

# Width of bar
ggplot(diamonds, aes(x=cut))+geom_bar(fill="red", color="blue",
width =0.3)#width in range (0, 1)

# transparency of th fill of bar
ggplot(diamonds, aes(x=cut))+geom_bar(fill="red", color="blue",
alpha=0.1)#alpha for transparency (0,1)
#Add labels
ggplot(diamonds, aes(x=cut))+geom_bar(fill="red",
color="blue")+labs(x="Diamonds Cut", y="Count", title="Frequencies
of cut")

# fill color to each bar as per cut
```

```
ggplot(diamonds, aes(x=cut))+geom_bar(fill="green")
ggplot(diamonds, aes(x=cut, fill=cut))+geom_bar()

#labels values to each bar by calculating count of each variable
group_by(diamonds, cut) %>% summarise(Freq=n())%>%
  ggplot(aes(x=cut, y=Freq))+
  geom_bar(stat="identity")+geom_text(aes(label=Freq))

# stat= identity is needed in above case pleas note this
group_by(diamonds, cut) %>% summarise(Freq=n())%>% ggplot(aes(x=cut,
y=Freq, fill=cut))+
  geom_bar(stat="identity")+
  geom_text(aes(label=Freq),vjust=-1, size=2, color="blue")#size to
change font and vjust for adjustment of position

#explore different size and positions
group_by(diamonds, cut) %>% summarise(Freq=n())%>% ggplot(aes(x=cut,
y=Freq, fill=cut))+
  geom_bar(stat="identity",width=0.9)+
  geom_text(aes(label=Freq),vjust=0.01, size=2.6, color="blue")

#plot quality of diamond with color(Biavriate two categorical)
View(diamonds)
ggplot(diamonds, aes(x=cut, fill=color))+geom_bar()  #stack bar
first var is cut, second is color
ggplot(diamonds, aes(x=cut, fill=clarity))+geom_bar()
ggplot(diamonds, aes(x=cut, fill=color))+geom_bar(position =
"dodge")  #grouped bar

#Use of facet_wrap
#We can get different plot in one frame  as per a categorical
variable
ggplot(diamonds, aes(cut, fill=cut))+geom_bar()+
  facet_wrap(~color)
ggplot(diamonds, aes(color, fill=color))+geom_bar()+
  facet_wrap(~cut)



ggplot(diamonds, aes(x=cut,fill=clarity))+
  geom_bar()+facet_wrap(~color)#for 3 categorical variables
```